

Syed Muhammad Rehan

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PROFESSIONAL SUMMARY

Software Engineer focused on production RAG systems, AI agents, NL2SQL, and cloud-deployed AI applications. Currently building enterprise AI solutions at Logic Powered Solutions, including a Watermarks Construction knowledge assistant on Azure AI Search/Microsoft Teams, a natural-language SQL agent for Azure SQL databases, and an autonomous QA agent using MCP, GPT-4o, and Playwright. Completed B.Sc. Software Engineering from FAST University, with a portfolio spanning multi-agent geospatial AI, LLM fine-tuning, retrieval pipelines, FastAPI backends, and client-facing delivery for international stakeholders.

EXPERIENCE

Software Engineer

Aug 2025 – Present

Logic Powered Solutions

F-5, Islamabad, Pakistan

- **Enterprise RAG Knowledge Assistant:** Designed and deployed a production assistant for Watermarks Construction using Azure AI Search, Azure OpenAI, Microsoft Copilot Studio/Teams, and SharePoint-indexed enterprise documents.
- **Advanced Retrieval Pipeline:** Implemented RAPTOR-style hierarchical retrieval with cluster summarization, OCR ingestion for scanned files, document chunking, vector search, and source-grounded answers.
- **Natural Language SQL Agent:** Built a Python NL2SQL agent that translates plain-English questions into optimized SQL across 3 Azure SQL databases, with schema-aware generation and readable responses.
- **Azure Deployment:** Deployed AI workflows through Azure Function App, Azure Bot Service, Microsoft Teams, and Azure SQL-backed enterprise infrastructure.
- **Autonomous QA Agent:** Developed an MCP + GPT-4o + Playwright testing agent that explores web applications, discovers user flows, writes test scripts, executes tests, and streams progress through SSE.
- **Client Communication:** Led requirement-gathering and progress discussions with Australian stakeholders through Microsoft Teams, translating user needs into deliverable AI behavior.
- **Backend Engineering:** Built and maintained Python/REST backend services for campaign workflows, targeting logic, authentication-adjacent flows, and third-party API integrations.

AI/ML Intern

Jun 2025 – Aug 2025

A Hamson

Rawalpindi, Pakistan

- Developed an internal RAG-based documentation assistant using LangChain, improving access to internal knowledge and reducing manual document lookup.
- Containerized Python services with Docker and supported Kubernetes-based deployment workflows for reproducible AI service environments.
- Supported database administration and web UI improvements with a focus on reliable data handling and practical usability.

EDUCATION

FAST University

Pakistan

B.Sc. Software Engineering

2021 – 2026

- **Final Year Project:** GeoVision, an AI-powered land-use planning system combining XGBoost environmental suitability, RAG-based zoning analysis, GIS spatial optimization, and a React/FastAPI planning interface.
- Relevant coursework: Artificial Intelligence, Machine Learning, Database Systems, Software Design and Architecture, Web Engineering, Operating Systems, Computer Networks.

Army Public School and College, Fort Road

Rawalpindi

FSc Pre-Engineering – 85%

2019 – 2021

PROJECTS

Enterprise RAG Assistant | *Azure AI Search, Azure OpenAI, Teams, SharePoint, Python* | [GitHub](#)

- Built a Teams-accessible knowledge assistant over SharePoint-indexed company documents with hybrid/vector retrieval, citations, and Azure-hosted bot infrastructure.
- Prepared production-readiness plan covering RAG quality tests, Azure Container Apps deployment, key rotation, conversation storage, and index freshness monitoring.

RAPTOR-Style Document RAG Pipeline | *Python, LangChain, ChromaDB, OCR, KMeans*

- Implemented recursive document loading, scanned-document OCR, chunk embedding, KMeans cluster summaries, and expanded retrieval for construction documents.
- Added caching and Streamlit-facing pipeline management to support repeatable indexing, retrieval experiments, and faster iteration.

Natural Language SQL Agent | *Python, Azure Function App, Azure Bot Service, Azure SQL*

- Converts business questions into schema-aware SQL across 3 Azure SQL databases and formats responses for non-technical users.
- Designed for Teams-accessible usage with guardrails around query construction, result explanation, and operational deployment.

QA Application | *Next.js, Node.js, GPT-4o, Playwright, MCP, SSE* | [GitHub](#)

- Built an LLM tool-use loop where the agent receives browser state, selects actions through Playwright MCP, discovers flows, writes tests, and executes them.
- Added real-time progress streaming, optional Swagger/OpenAPI guidance, generated test file management, and path-safety checks.

GeoVision | *React, FastAPI, XGBoost, ChromaDB, GeoPandas* | [GitHub](#)

- Designed a multi-agent planning workflow with an Environment Agent for suitability scoring, a Zoning Agent for RAG-based compliance context, and a Spatial Agent for GIS geometry generation/optimization.
- Implemented FastAPI job orchestration with status polling, stage tracking, merged outputs, and a map-centric React interface for parcel selection and plan comparison.

LLM Fine-Tuning Pipeline | *Python, Unsloth, QLoRA/LoRA, Hugging Face, Ollama* | [GitHub](#)

- Built a reproducible script pipeline for setup, model loading, dataset preparation, LoRA training, adapter export, Ollama packaging, and interactive inference.
- Documented Windows/WSL2 GPU setup, 4-bit model loading, VRAM controls, dataset expectations, and fallback export paths.

B.M.O Interactive 3D AI Character | *Next.js, React Three Fiber, FastAPI, ChromaDB, Whisper* | [GitHub](#)

- Created a 3D conversational AI character with state-driven facial expressions, speech-to-text input, vector memory, OpenAI responses, and voice conversion.
- Integrated frontend 3D interaction with backend AI endpoints for text generation, speech synthesis, memory retrieval, and conversation state.

TECHNICAL SKILLS

AI/ML and RAG: LangChain, LangGraph, RAG pipeline design, RAPTOR-style retrieval, vector search, ChromaDB, prompt engineering, LLM fine-tuning (QLoRA/LoRA), OpenAI/Azure OpenAI APIs, Whisper, XGBoost, embeddings, OCR

Agents and LLM Apps: Tool calling, MCP, autonomous browser agents, NL2SQL, multi-agent orchestration, structured outputs, retrieval evaluation awareness, conversational memory

Cloud and Azure: Azure AI Search, Azure Function App, Azure Bot Service, Microsoft Copilot Studio, Azure SQL Database, Azure Container Apps awareness, Azure Portal

Backend and APIs: Python, FastAPI, Flask, Node.js, Express, REST APIs, JWT, async workflows, WebSockets, SSE, SQL query optimization

DevOps, Data, Other: Docker, Kubernetes, Git/GitHub, Linux/Ubuntu, MySQL, Azure SQL, MongoDB, SQLite, GeoPandas, slight frontend exposure

Testing and QA: Playwright, pytest, generated test suites, browser automation, MCP-based tool execution

ADDITIONAL INFORMATION

Languages: English (professional; client-facing meetings with Australian stakeholders), Urdu (native)

Communication: Technical demos, requirement gathering, progress reporting, and translating AI system behavior for non-technical users

Career Focus: AI Engineer, RAG Engineer, Agentic AI Engineer, GenAI Engineer, Python AI Engineer, Full Stack AI Engineer

Availability: Open to full-time AI/ML engineering roles and freelance AI/RAG projects